

Keeper tier-gate — Two-Route Over-Determination Scan v0.1 (Grace) + Quasi-Eigentone Framework v0.1 (Lyra). Two-Route Scan: PASS at STRUCTURAL with null-model condition before any Casey-named-principle elevation; 8-18 cluster finding is the central data; magic-126 product-only is honest asymmetry.

Quasi-Eigentone: PASS at FRAMEWORK with two v0.2 sharpening notes (neutrino mass-eigenstate granularity; V-compound coproduct burden).

Keeper

2026-05-30 Saturday

Tier-gate — Two-Route Scan v0.1 + Quasi-Eigentone v0.1

1. Grace Two-Route Over-Determination Scan v0.1

Verdict: PASS at STRUCTURAL. Computational method rigorous; results honest (8/39 BOTH, not universal); cluster finding (8-18 = SM-gauge dimensional content) is the central structural data; magic-126 product-only asymmetry is genuine substrate information.

Strengths: - Method rigorous (Python enumeration of all $1/2/3$ -prim sums + products + simple powers). - Results honest: 8 BOTH (~20%), 4 sum-only, 18 product-only, 11 NEITHER. NOT a universal pattern; explicit partial finding. - Trivially-definitional sum-routes ($5=\text{rank}+N_c=n_C$; $7=\text{rank}+n_C=g$) called out and excluded from substantive over-determination count. - $12 = 4$ -route over-determination flagged for Lyra theory engagement (n_C+g ; $\text{rank}+N_c+g$; $\text{rank}\cdot C_2$; $N_c\cdot\text{rank}^2$) — strongest substrate-arithmetic finding. - Cluster reading honest: doubly-over-determined zone (8-18) coincides with SM-gauge dimensional content; specific, not vague. - Exception structure (magic-126 product-only; $N_{\text{max}}=137$ NEITHER; magic-82/184 NEITHER; SM-26 NEITHER) is treated as substrate information, not noise.

Conditions for advancement:

- N1 — Null-model test before Casey-named-principle elevation: Section H proposes two candidate framings (higher-bar “Substrate Doubly-Over-Determination Principle” + lower-bar “Substrate-Arithmetic SM-Gauge Coincidence”). Before either advances to Casey-named candidacy, a Toy-1543-style null-model is required: how often does substrate-arithmetic (small primaries $2/3/5/6/7$ with sums/products/powers) hit a random small-integer set in the 1-50 range? The hit-rate context determines whether 8/39 is structurally meaningful or expected by counting. Until that null-model lands, the principle stays as “structural observation,” not Casey-named.
- N2 — Cluster claim wording: Section C says “the substrate over-determines exactly the SM’s load-bearing gauge counts via two structurally-independent arithmetic routes” — the “exactly” is too strong (only 8/18 integers in 8-18 are BOTH; e.g., 11, 13, 17 are not). Suggest v0.2 reword to “substrate doubly-over-determines a cluster of SM-relevant integers in 8-18, including SM gauge content (8 gluons, 12 total gauge dim).”

Tier disposition: - Method + Results (Sections A+B): RIGOROUS (computational, verified) - Cluster reading (Section C): STRUCTURAL observation - Predictions/falsifiers (Section E): FRAMEWORK at substrate-arithmetic level - 3-route / 4-route findings (Section F): STRUCTURAL — flag to Lyra for bulk-color theory engagement legitimate - Candidate Casey-named principles (Section H): CANDIDATE, blocked on N1 null-model

Status: PASS at STRUCTURAL; advances to candidate; null-model required before principle elevation. The 4-route over-determination of 12 (SM total gauge dim) is the strongest substrate-arithmetic finding to date and is legitimately flagged to Lyra for bulk-color (#418) theory work.

2. Lyra Quasi-Eigentone Framework v0.1

Verdict: PASS at FRAMEWORK with two v0.2 sharpening notes. The structural distinction (true vs quasi vs eigentone-in-vacuum) is engine-level coherent; generalizes Elie E3 β -decay cleanly; honest about open quantitative work.

Strengths: - Engine-level mechanism (Green coproduct + grading conservation) provides clean structural mechanism for decay channels. - Per-particle classification consistent: stable particles ($e, \nu, \gamma, p, \text{gluon}$) = sector bottoms; unstable ($\mu, \tau, n, W, Z, H, \text{hadrons}$) = excited states with coproduct decay. - Generalization from E3 β -decay to full SM is structurally appropriate. - Conservation = grading on the Hopf algebra (Q, B, L automatic) is the right substrate-mechanism for the conservation laws. - Open items (quantitative decay rates, branching ratios, CP-violation) explicit and tier-honest. - Movie program V0/V3 narrative-spine tie-in is appropriate (not load-bearing). - Cal #27 self-applied: “this v0.1 places the structural DISTINCTION ... it is NOT a full derivation of decay rates.”

Notes for v0.2 sharpening (Cal cold-read recommended):

- N1 — Neutrino classification granularity: Section 3 classifies “neutrinos” as TRUE EIGENTONE collectively. More precisely: only the lightest neutrino mass eigenstate (ν_1) is strictly a true eigentone (ground state in its sector). ν_2 and ν_3 are quasi-eigentones in the mass-eigenstate basis, but they DO NOT DECAY — they oscillate via PMNS mixing. The framework should distinguish “no decay channel” (true eigentone behavior) from “oscillation without decay” (a separate category that ν_2/ν_3 inhabit). Suggest v0.2 adds “OSCILLATING” or treats ν_2/ν_3 as a sub-class. The classification “neutrinos all true eigentone” is structurally correct under the “no decay” criterion but loses the PMNS structure.
- N2 — V-compound K-type coproduct burden: Section 3 classifies pion, K-mesons, Λ baryon, and “other hadrons” as QUASI in “V-compound” K-types. The framework asserts these have Green-coproduct decay channels, but the V-compound K-type structure is not yet explicit at engine level. Engine v0.3 §6/§7 (Elie’s Drinfeld double + bulk-color algebraic) hints at the mechanism but doesn’t yet construct it. Suggest v0.2 marks V-compound coproduct as “structural assertion pending engine v0.3 closure,” not derivation. The fundamental K-types ($V_{(1/2,1/2)}$ lepton; $V_{(1,0)}$ photon; $V_{(1,1)}$ gauge bulk) have clean engine images; the composite/compound K-types do not yet.

Tier disposition: - Engine-level mechanism (Green coproduct + grading conservation): RIGOROUS (engine v0.1/v0.2 base) - Structural distinction (true/quasi/eigentone-in-vacuum): FRAMEWORK — well-motivated, structurally clean - Per-particle classification: FRAMEWORK with sub-tier caveats: - Leptons (e, μ, τ, ν as Dirac per F5): FRAMEWORK — clean K-type - Photon ($V_{(1,0)}$): FRAMEWORK — clean - Gluon (eigentone-in-vacuum): FRAMEWORK — confinement framing acceptable - Proton ($V_{(1,1)}$ bulk $k=6$): FRAMEWORK — consistent with F5 (no GUT proton decay) - W, Z, Higgs (gauge $V_{(1,1)}$): FRAMEWORK — coproduct mechanism in place - Hadrons (V-compound): STRUCTURAL — coproduct mechanism asserted, awaits engine v0.3 V-compound layer - Open items (quantitative decay rates, branching ratios, CP-violation): explicitly OPEN, multi-week

Status: PASS at FRAMEWORK; ν_2/ν_3 granularity + V-compound coproduct burden are the v0.2 sharpening items. Once Elie’s engine v0.3 lands with V-compound layer, the hadron classification consolidates from STRUCTURAL to FRAMEWORK.

3. Routed handback

- Grace: N1 null-model — design a Toy that asks “given a random small-integer set of size 39 in range 1-200, how often does substrate-arithmetic (sums + products + powers of {2,3,5,6,7}) doubly-cover ≥ 8 of them?” Once filed, the 8/39 finding gets context. Then candidate Casey-named principle can advance to cold-read.
- Lyra: N1 ν -eigenstate granularity refinement (mass eigenstates ν_1 vs ν_2/ν_3 ; oscillation not decay) + N2 V-compound coproduct burden marked as “pending engine v0.3” — both go into Quasi-Eigentone v0.2.
- Cal: cold-read queue extension — Two-Route Scan v0.1 (null-model question), Quasi-Eigentone v0.1 (structural framework cold-read).
- Elie: engine v0.3 V-compound layer (§7 + new §8?) is the next consolidation that closes Quasi-Eigentone v0.2 N2.

4. Queue continuation

Pulled and dispositioned today (Saturday afternoon): 5 items 1. Lyra bulk-color v0.4 (route commitment C+D) — FRAMEWORK with C1+C2 2. Elie engine v0.2 audit support v0.2 (15 toys 3612-3626) — PASS; §6+§7 STRUCTURAL pre-reserved 3. Lyra T1 falsifier v0.1 (lepton count 24) — PASS with v0.2 sharpening 4. Grace Two-Route Scan v0.1 (8/39 BOTH cluster) — STRUCTURAL with N1+N2 5. Lyra Quasi-Eigentone Framework v0.1 — FRAMEWORK with N1+N2

Pulling next: - Lyra spectral score theoretical layer v0.1 (Lyra 10:00) - Lyra Vol 16 Ch1 §A_sub-Hall Dictionary v0.1 (Lyra 10:01) - Engine v0.3 consolidation doc RE-AUDIT (pending Elie filing) - Cal cold-reads as they land

— Keeper. Two more tier-gates filed; both advance to candidate-status with explicit conditions; team queue continues. Five afternoon dispositions total; queue holding sustainable cadence.